

Contact

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Top Skills

Sonarqube
Google BigQuery
Jenkins

Certifications

Agile Certified Product Manager and
Product Owner Credential (ACPM)
The Data Scientist's Toolbox

Honors-Awards

Petroleum Technology Development
Fund Scholarship
Codeplus Software Academy
Internship Program Excellence
Award
Demz Analytics Data Science Track
Internship Program Excellence
Award

ayodeji ajayi

Python | Senior Backend Engineer | AI/LLM systems | Data scientist |
Distributed systems | Fintech | Financial Data Analyst
Kosofe, Lagos State, Nigeria

Summary

I am a seasoned Python Back-End Engineer with over 8 years of professional experience in software engineering, specializing in building scalable, high-performance server-side applications and APIs. My core expertise lies in designing robust back-end architectures, optimizing database schemas, and implementing system automation using Python and modern frameworks. I bring strong experience in containerized environments, CI/CD pipelines, and orchestration tools, supporting reliable deployments and production stability. In addition, I have a solid background in data science and engineering, which enables me to integrate data-driven features and analytics into back-end systems. I thrive in Agile environments and am passionate about writing clean, maintainable code that drives efficiency, business growth, and informed decision-making.

Having majorly worked in Financial institutions and AI-driven organizations, I have built diverse services and models such as Recommendation systems, Asset/Investment management systems, Data mining Applications, Social media monitoring applications(for intelligence gathering), Computer vision applications, Natural Language models, Machine Learning models, Document processing applications, Health management systems, E-commerce applications, Artisan management applications, etc.

Experience

Demz Outsourcing Solutions
Senior AI / Backend Engineer & Architect
January 2025 - Present (1 year 8 months)
Lagos

❖ Architected and built AI-assisted backend systems integrating local LLMs (LLaMA3, Gemma via Ollama) into engineering and e-commerce workflows —

reducing development cycle time by 40% through intelligent code scaffolding and automated API generation.

- ❖ Designed agentic workflows automating engineering operations including code scaffolding,

API generation, debugging assistance, and developer productivity tasks — improving team velocity by 35%.

- ❖ Established architecture patterns for retrieval-augmented generation (RAG) workflows, embedding-based semantic search, and contextual AI interactions across the platform using vector databases.

- ❖ Led end-to-end delivery of an AI-assisted e-commerce platform with React/Redux frontend and Python microservices backend, serving as technical and AI architect.

- ❖ Built prompt engineering frameworks and LLM orchestration layers for reliable, observable AI behavior in production.

- ❖ Managed production systems including monitoring, logging, debugging, and performance optimization across distributed AI services.

Andela

Senior Software Engineer

December 2022 - Present (3 years 9 months)

Lagos State, Nigeria

I currently work in my capacity as a Senior Software Engineer with Wynden Startk LLC dba GQR Global Markets, an Andela client, under the Nebula innovation team.

Nebula.io

Senior Software Engineer

January 2023 - December 2024 (2 years)

United States

- ❖ Responsible for developing, testing, and architecting scalable containerized microservices using Python, Flask framework, Docker, etc.

- ❖ Participate in the Agile development process to quickly and efficiently provide deliverable solutions.

- ❖ Review team members' code to ensure it meets all requirements and standards.

Asset & Resource Management Holding Company (ARM HoldCo).

Senior Backend Engineer(Python) | Data Scientist

January 2022 - February 2023 (1 year 2 months)

Lagos, Nigeria

- ❖ Building data pipelines and APIs, optimizing the data flow, and preparing data for actionable intelligence. Automating manual processes, optimizing data delivery, and re-designing infrastructure for greater scalability. Analyzing acquisition, operational efficiency, and other critical business performance metrics. Contributing to the design, architecture, and development of products. Translating business requirements into technical functionalities and code. Containerizing applications, setting up workflows, adding containers to an existing cluster, and configuring servers.
- ❖ Shaped the backend for ARMONE, an application that aggregates customer's investment portfolios, converts transactional data into actionable insights for customer acquisition, operational efficiency and other key business performance metrics, etc.
- ❖ Developed KYC (Know Your Customer) service; a computer vision solution that validates client's document and identity prior to onboarding.
- ❖ Built the backend of EASYWILL, a web API that manages client Estate details(valuable assets such as Stocks/Shares, Investments, Life policies, Retirement savings accounts, Real estate, Bank accounts, Gifts, Financial assets, Bequests of Real Property, etc.), Estate Executors, Residuary Estate(proportion of sharing assets) as will.
- ❖ Mentoring and guiding less experienced members of the engineering team.
- ❖ Conducting vulnerability and penetration tests while participating in agile development and processes, sprint planning, sprint retrospective, sprint review, backlog refinement, etc.

Demz Analytics

Python Developer, Machine Learning Engineer and Data Scientist

May 2019 - January 2022 (2 years 9 months)

London, England, United Kingdom

- ❖ Develop distributed systems using Microservice Architecture.
- ❖ Translate business requirements into technical functionalities and software codes
- ❖ Responsible for building several NLP-based applications for extracting and analyzing document information, document clustering, language modelling, text processing, text classification, topic modelling, sentiment analysis etc.
- ❖ Responsible for building several Computer Vision based applications for identity verification, object detection, object tracking, image manipulation, image classification, image segmentation, and converting news feeds into videos.

- ❖ Finished up a recommender solution that analyzes users' interests based on the entities in the videos interacted with, and their weighted actions to offer recommendations that suites their interests.
- ❖ Setup Elasticsearch Database (4 node replicas set database for powering search and analyzing big data in a matter of milliseconds), an interface using Kibana to allow easy interaction with the database, and Logstash to monitor and migrate data from the mongo database. Created web API that houses all necessary utilities to carry out actions like fuzzy searches, partial searches, sorting, pagination, search-as-you-type, save, retrieve and delete data from the database.
- ❖ Configure servers for container orchestration, container distribution across nodes, load balancing, reverse proxy, CI/CD and several other automation in both staging and production environments.
- ❖ Responsible for building several data crawling services to gather top trending articles based on the interactions like retweets, replies and like actions.
- ❖ Conduct vulnerability assessment and penetration tests on services built by my team members to avoid information disclosure and credentials harvesting from error handling, enforcing rate limiting and also ensuring that identified concerns are adequately remediated before products are released.

Teknohub Solutions Management Limited

Back End Developer

November 2016 - December 2018 (2 years 2 months)

ikosi-ketu

- ❖ Responsible for building several SAAS applications.
- ❖ Developed a health management solution that handles customer onboarding, and appointment booking with hospitals, diagnostic centres, HMOs, doctors, and emergency responders.
- ❖ Contributed to the development of EasyReg solution; an application used by pension fund administrators to onboard customers and capture their biometric information.
- ❖ Mentored junior developers and interns.

Chevron

Internship Trainee

May 2013 - October 2013 (6 months)

Port Harcourt, Rivers State, Nigeria

Education

University of Benin

Bachelor of Engineering - BE, Petroleum Engineering · (2009 - 2014)